American Computer Science League

2025-26 Finals ● Short Problems ● Intermediate/Classroom Division

1. Boolean Algebra

How many ordered triples make the following Boolean expression TRUE?

$$\overline{A + B} * C + \overline{A}(B + C) + \overline{AC} * B$$

A. 8
B. 6
C. 4
D. 2
E. None of the above

2. Boolean Algebra

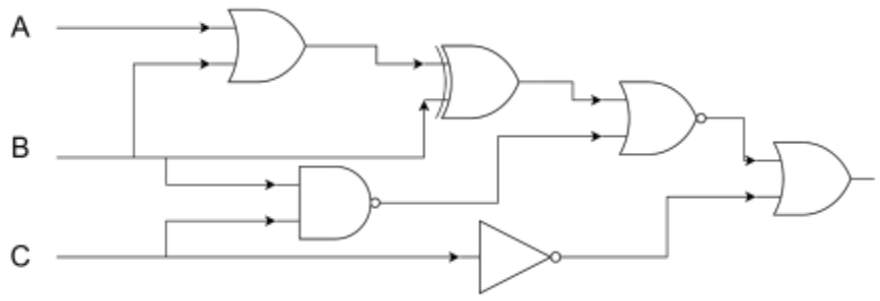
Simplify the following Boolean expression:

$$(A + \overline{B})(\overline{\overline{B} + \overline{C}}) + (\overline{A + \overline{B\,C} + C(\overline{\overline{A} + B})})$$

A. $\overline{B}C$
B. BC
C. $B\overline{C}$
D. ABC
E. None of the above

3. Digital Electronics

What Boolean expression is a direct translation of the following digital circuit?



A. $\overline{((A + B) \oplus B) + \overline{\overline{BC}} + \overline{C}}$
B. $\overline{((A + B) \oplus B) + BC + C}$
C. $\overline{((AB) \oplus B) + \overline{\overline{B + C}} * \overline{C}}$
D. $\overline{((AB) + B) + \overline{\overline{BC}} \oplus \overline{C}}$
E. None of the above

4. Bit-String Flicking

Evaluate the following expression for 8-bit strings and express the answer in octal:

(LCIRC-2 $A5_{16}$) OR (RSHIFT-1 10010101_2)

A. BE
B. 176
C. 236
D. 336
E. None of the above

5. Bit-String Flicking

Determine what bit strings make the following expression TRUE:

(LCIRC-3 ((RSHIFT-1 01111) OR (RCIRC-2 X) AND (LSHIFT-1 11111))) = 11001

A. **110
B. ***00
C. *1**1
D. *0*1*
E. None of the above

6. Recursive Functions

Find $f(7)$, given:

$$f(x) = \begin{cases} 2x+1 & \text{if } x < 3 \\ 2 \cdot f(x-2) - f(x-1) + 3 & \text{if } x \geq 3 \end{cases}$$

A. -12
B. 19
C. 9
D. 2
E. None of the above

7. Recursive Functions

Find $f(5, 7)$, given:

$$f(x,y) = \begin{cases} f(y-3, x+1) - 3 & \text{if } x+y \geq 8 \\ 2 \cdot f(x-2, y+1) + 2 & \text{if } 5 \leq x+y < 8 \\ x+y & \text{if } x+y < 5 \end{cases}$$

A. 22
B. 19
C. 16
D. 13
E. None of the above

8. Prefix/Infix/Postfix Notation

Define:

a & b = $a^2 - b$
a % b = [(a+b) / 2] where [x] is the greatest integer less than or equal to x

Evaluate the following postfix expression if all numbers are single digits:

4 3 % 3 3 & 2 1 & * 3 2 ^ / + 8 2 3 % / −

A. 0
B. 1
C. 2
D. 3
E. None of the above

9. Prefix/Infix/Postfix Notation

Convert the following infix expression to prefix:

$$\frac{(A - B^3 + C)^2}{(\frac{A+C}{B})}$$

A. /^+−A^B3C2/+ACB
B. /^−+A^B3C2/+ACB
C. /^+−^/+AB3C2ACB
D. /^+−2C^3B2A/+BCA
E. None of the above

10. Computer Number Systems

What is the average of the octal numbers 152, 271, and 713? Express your answer in base 10.

A. 250
B. 750
C. 1000
D. 1356
E. None of the above

11. Computer Number Systems

What is the value of x?

$$x_8 = (ABC_{16} + 864_{16}) / 10_{16}$$

A. 1320
B. 462
C. 262
D. 132
E. None of the above

12. Data Structures

Build a binary search tree for:

FIFAWORLDCUP

What is the external path length?

A. 34
B. 54
C. 58
D. 75
E. None of the above

13. Data Structures

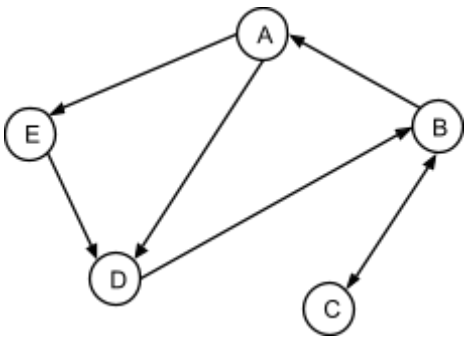
If the string "ACSL program" is typed into your word processor followed by the following Undo (↩) and Redo (↪) commands that remove the last character typed or insert the last character removed, what is the resulting string?

↩ ↩ ↩ ↩ ↩ ↩ ↪ ↩ ↩ ↪ ↩ ↩ ↪ ↪ ↪ ↩ ↩ ↩ ↩ ↩

A. ACS
B. ACSL
C. ACSL p
D. AC
E. None of the above

14. Graph Theory

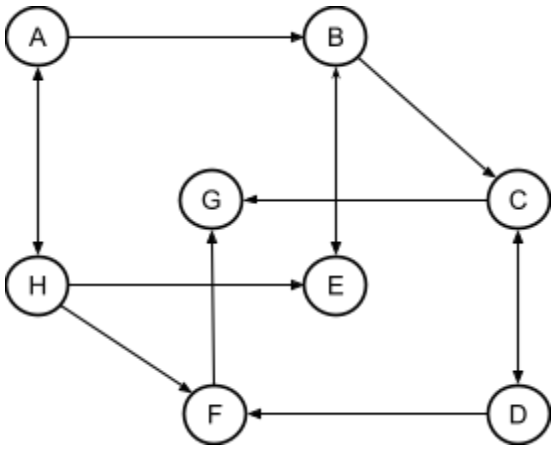
Given the directed graph below, which vertex has the most paths of length 3 originating from it?



A. A
B. B
C. C
D. D
E. None of the above

15. Graph Theory

How many cycles involving more than 2 vertices are there in the following directed graph?



A. 1
B. 2
C. 3
D. 4
E. None of the above

16. What Does This Program Do?

Given that array **arr** has **n** items, the following program uses the Selection Sort to put values in the array into ascending order. Fill in the blank line so that it works correctly, given that the first item is stored in location 0.

```
for j = 0 to n-1
    min = j
    for k = j+1 to n-1
        if arr(k) < arr(min) then
            [          ]
        end if
    next k
    t = arr(j)
    arr(j) = arr(min)
    arr(min) = t
next j
```

A. `k = min`
B. `arr(min) = arr(k)`
C. `min = k`
D. `min = j`
E. None of the above

17. LISP Programming

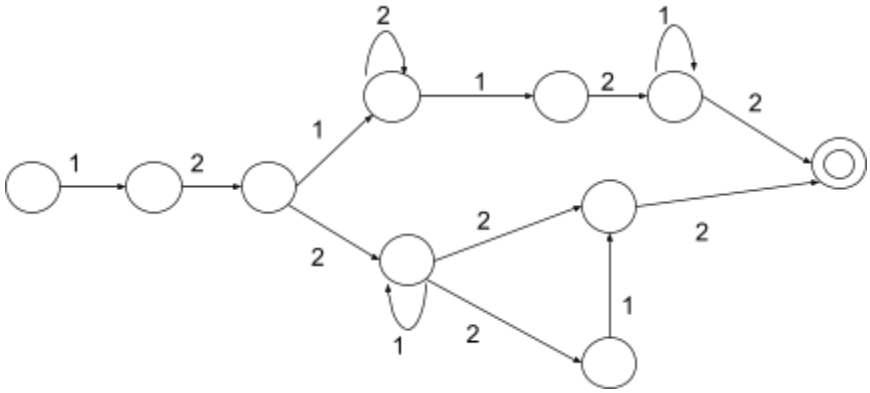
Evaluate the following LISP statements:

```
(SETQ X '((SUB 7 2) 5 (ADD 3 1)))
(SETQ Y '((MULT 4 3 5) (DIV 2 1) (6 (3 4) 2)))
(EXP (EVAL (CADDR X)) (EVAL (CADR Y)))
```

A. 4
B. 8
C. 16
D. 64
E. None of the above

18. FSAs and Regular Expressions

Which of these strings are accepted by the following FSA?



a. 12122222	b. 1222112	c. 121122
d. 1221111111212	e. 12122222122	f. 122111212

A. a,b,d,f
B. c,d,e,f
C. a,c,d,e
D. b,c,d,f
E. None of the above

19. FSAs and Regular Expressions

How many of the following strings are accepted by the following regular expression?

[A-C] [a-hj-np-z]* . [1-9] [0-9]*

Bfractals#8000	Afib:11235	Calculate!0456
Carfun@1089	Addsum+6789	Bravado012

A. 2
B. 3
C. 4
D. 5
E. None of the above

20. Assembly Language

What is the output when the following program is executed?

```
A       DC          13
B       DC          21
NUM     DC          7
TOP     LOAD        NUM
        SUB         =11
        BE          STOP
        ADD         =12
        STORE       NUM
        LOAD        A
        ADD         B
        STORE       C
        LOAD        B
        STORE       A
        LOAD        C
        STORE       B
        BU          TOP
STOP    PRINT       B
        END
```

A. 34
B. 55
C. 89
D. 144
E. None of the above